Supervised Bayesian Matrix Factorization Identifies Prognostic Gene Expression Programs

Project Update

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Agenda

Background · The PDAC prognostic problem · why supervise the factorization

Method updates · The joint model & explicit priors · non-parametric baseline hazard · the ELBO & variational family · CAVI updates & convergence · preprocessing · selecting K: ARD shrinkage + a consensus of fit/prediction metrics

Results · The K_{init} consensus sweep · gene programs & survival · joint model vs. the unsupervised two-step, including as survival signal strength varies

Impact & Future Directions

Roadmap for the next ~15-20 minutes. K selection works in two stages: a consensus of ELBO, BIC, log-likelihood, and cross-validated external C-index picks K_init, and ARD shrinkage then determines which of those factors are actually survival-active (K_eff). The projection predictor (YF) is the only model presented.

Background

Background: PDAC needs molecular prognosis

- **Five-year survival $< 12\%$** — among the most lethal solid tumors
- Most patients diagnosed late; **no widely adopted molecular stratification** at diagnosis
- Bulk expression mixes malignant, stromal, and immune signals — prognostic structure is interwoven and often **low-variance**
- **Goal:** recover *gene expression programs (GEPs)* whose activation predicts survival, and validate them on **independent** cohorts
- A GEP is a coordinated set of genes (a column of F) whose patient-level activation (a column of L) we test for survival association

> **ℹ** Why model expression and survival *jointly*?
>
> The common pipeline is **two-step**: run unsupervised factor analysis (PCA / EBMF), then test factors for survival *post hoc*.
>
> - Unsupervised axes track the **largest expression variance** — often batch or housekeeping signal

- A program explaining little variance but strongly prognostic gets **missed**
- **Supervision** steers factors toward survival-relevant structure *during* estimation

A program explaining 15% of variance may be pure batch effect; one explaining 3% may cleanly separate survivors. Supervision prioritizes the second.

One background slide. The clinical motivation is unchanged. The methodological motivation — joint over two-step — is the thread of the whole talk, and I back it with both a synthetic comparison and a real-data external comparison against the unsupervised two-step (EBMF → Cox) baseline.

Methods

The model — two coupled components

$$\underbrace{Y = LF^{\top} + E, \quad E_{ij} \sim N(0, \tau_j^{-1})}_{\text{genomics: low-rank+gene-specific noise}} \qquad \underbrace{h_i(t) = h_0(t)\exp(\eta_i)}_{\text{Cox proportional hazards}}$$

Quantity	Dim	Meaning
Y	$n \times p$	expression matrix
L	$n \times K$	patient program-activation scores
F	$p \times K$	gene weights per program
β	$K \times 1$	survival (log-HR) coefficients
τ_j	scalar	gene-specific noise precision
η_i	scalar	patient log-risk score, $(Y_i F)\beta$

Shared F couples the two

The **same** gene-program weights F drive both the low-rank reconstruction of Y **and** the survival risk score $\eta = (YF)\beta$. A program is **prognostic only if its** $\beta_k \neq 0$; otherwise it explains expression alone.

Non-parametric baseline hazard

$h_0(t)$ is left **unspecified** — it cancels in the Cox partial likelihood. No Weibull / exponential shape assumption; only proportional hazards.

Two equations, one shared latent structure. Every symbol is defined in the table, including the risk score eta, which is now always the projection score (YF) . The baseline hazard is non-parametric, so h0(t) drops out of the partial likelihood and we make no parametric shape assumption.

The model: projection predictor and explicit priors

Projection predictor — $\eta = (YF)\beta$

- Risk uses the **direct genomic projection** YF, not a re-derived patient score
- **Exact** out-of-sample scoring: $\eta_{\text{new}} = (Y_{\text{new}}F)\beta$ — the *identical* formula at train and test, no train/test score mismatch
- L is fit for the genomics reconstruction only; it does not enter the survival likelihood

> Why the projection form
>
> Leakage-free **exact** external scoring, with no OLS re-projection step for new patients — the property that makes external validation on held-out cohorts a clean apples-to-apples comparison.

Priors — every block is explicit. $q(\cdot)$ denotes the **variational (approximate posterior)** for each block, fit by empirical Bayes at every update:

- $q(L_{\cdot k})$, $q(F_{\cdot k})$ — **point-exponential** (non-negative spike-and-slab): $g(\theta) = \pi_0\delta_0 + (1-\pi_0)\,\text{Exp}(\lambda),\ \theta \geq 0$
- $q(\beta_k)$ — **normal** (Gaussian slab, no spike): $g(\beta) = N(0, \sigma^2)$
- $q(\tau_j)$ — **closed-form MLE**: no EBNM, no prior tuning

Sparsity π_0, scale λ, and slab variance σ^2 are all estimated from the data (`ebnm`) at each step. The **normal prior on** β, not a spike-and-slab, is what makes ARD-based K selection possible (next: *selecting* K) — shrinkage toward zero, not a hard on/off switch.

The projection predictor (YF) scores new patients with the identical formula used at train time — no mismatch, no re-derivation. Priors are explicit: non-negative point-exponential on L and F, a Gaussian (normal) prior on , and a closed-form MLE for . The normal prior on   is worth flagging here because it's exactly the mechanism ARD-based K selection relies on next.

The objective: ELBO & variational family

Mean-field variational family — each block factorizes independently:

$$q(L, F, \beta, \tau) = \prod_{k=1}^{K} q(L_{\cdot k}) \prod_{k=1}^{K} q(F_{\cdot k}) \prod_{k=1}^{K} q(\beta_k) \prod_{j=1}^{p} q(\tau_j)$$

The **ELBO** (evidence lower bound) is the objective CAVI maximizes — a lower bound on the log marginal likelihood. For the projection model:

$$\text{ELBO} = \underbrace{E_q[\mathcal{L}_{\text{gen}}]}_{\text{expression fit}} + \underbrace{\lambda\, E_q[\mathcal{L}_{\text{Cox}}]}_{\text{survival fit}} - \underbrace{\textstyle\sum_{\text{blocks}} D_{\text{KL}}(q \,\|\, \text{prior})}_{\text{prior regularization}}$$

- **Genomics fit** $E_q[\mathcal{L}_{\text{gen}}] = \sum_j E_q \log p(Y_{\cdot j} \mid L, F_j, \tau_j)$ — Gaussian reconstruction of Y by $LF^\top$
- **Survival fit** $E_q[\mathcal{L}_{\text{Cox}})]$ — a **2nd-order Taylor** expansion of the Cox partial log-likelihood in $\eta = (YF)\beta$, making the survival term conjugate to the EBNM updates

- **KL term** penalizes each posterior q away from its prior — the adaptive shrinkage
- The **shared** F enters both likelihood terms — genomics reconstruction ($Y \approx LF^\top$) and survival risk ($\eta = (YF)\beta$) — that coupling *is* the supervision

- The ELBO **balances** expression and survival automatically — **no tuning parameter** α ($\lambda = 1$ in the recommended fits)

  What the ELBO is *for*: it is the **fitting objective** — CAVI maximizes it, and a fit is declared converged when it stops improving. It is also **one of four criteria** — alongside BIC, log-likelihood, and cross-validated external C-index — used to choose K_{init} (next: *Selecting* K).

The variational family is mean-field and factor-wise. The ELBO has three pieces: Gaussian genomics reconstruction, the Cox survival term via a second-order Taylor expansion that keeps the updates conjugate, and a per-block KL penalty that is the adaptive shrinkage. The shared F in both likelihood terms is the supervision. The ELBO is one of the four K_init-selection criteria alongside BIC, log-likelihood, and cross-validated external C-index.

Inference: factor-wise CAVI

Coordinate ascent (Gauss–Seidel) — per outer iteration, for each factor k:

$$\beta_k \;\rightarrow\; L_{\cdot k} \;\rightarrow\; F_{\cdot k}, \qquad \text{then } \tau \text{ once per sweep}$$

- Each update is an **EBNM** sub-problem: pseudo-data $x = B/A$, scale $s = 1/\sqrt{A}$, then `ebnm` returns the posterior mean **and** variance
- β_k has precision governed by the projection score's own variance across patients, so a factor with **no survival signal** shrinks $\beta_k \to 0$ on its own — the mechanism ARD relies on
- The empirical-Bayes prior $\hat{g}$ is **re-estimated every update** $\rightarrow$ adaptive shrinkage per factor / per gene; deflate the residual after each factor

Convergence — in plain English. After a 5-iteration burn-in, stop when the **ELBO stops improving** — its relative change between iterations falls below 10^{-5}:

$$\frac{\left|\mathrm{ELBO}^{(t)} - \mathrm{ELBO}^{(t-1)}\right|}{\left|\mathrm{ELBO}^{(t-1)}\right|} < 10^{-5}$$

- The ELBO is the **single objective** being maximized, so this directly detects a stationary point of the variational optimization
- Mean absolute changes in L and β are tracked alongside as **diagnostics** (both settle below 10^{-3} at convergence)

> **!** Risk-direction sign correction
>
> Cox concordance is invariant to a global sign flip, so an unconstrained fit can land at $C < 0.5$. **Fix:** orient each program against the **training** concordance, then apply the *same* orientation at test. (Used to define program activation in the survival results.)

CAVI updates one factor at a time in Gauss-Seidel order; each block reduces to an EBNM call with pseudo-data x=B/A and scale s=1/sqrt(A). Convergence requires both L and beta to settle below 1e-3 in MEAN absolute change; averaging avoids a single oscillating factor blocking the stop. The sign correction orients each program by training concordance and reuses it at test.

Preprocessing is crucial: per-platform normalization

The $p \gg n$ scaling problem. Merging platforms (RNA-seq, microarray, proteomics) leaves each gene on a different scale. The survival precision for program k is

$$A_k = \sum_i w_i\,(YF)^2_{ik},$$

where w_i = the patient's **Cox working weight** (curvature from the survival Taylor step) and $(YF)_{ik}$ = patient i's **projection score** on program k. Without correction, between-platform variance inflates (YF) unevenly, **swamps** A_k, and the survival coefficients **collapse to** $\beta \to 0$.

The fix — normalize *per platform, before merging*:

1. Log2 transform (RNA-seq only)
2. **Per-cohort gene selection** (combined rank of mean × variance), top-3000 per cohort, then **intersect** → **2064 genes** in the merged training matrix
3. **Per-platform z-standardization** (each cohort centered/scaled on its own)
4. Row-bind into the merged training matrix

! Skip per-platform normalization → collapses

In an 18-configuration benchmark, **10 of 12** pipelines that do **not** z-standardize per platform before merging drive $\beta \to 0$.

No leakage

Normalization is fit **within each training cohort, before** any split; held-out cohorts get the *same* per-cohort recipe at scoring time.

The single most important practical finding since April. A_k is the survival precision for program k. Without per-platform normalization the projection scales blow up across platforms, A_k is dominated by between-platform variance, and beta collapses. The fix is per-platform z-standardization before merging, which leaves 2064 genes. 10 of 12 alternatives collapse. Normalization is fit per training cohort before any split — no leakage.

Selecting the number of programs: a two-stage framework

Stage 1 — K_{init}, by a consensus of four fit/prediction criteria. Fit at several K_{init}; compare ELBO, BIC, an ELBO-style joint log-likelihood, and cross-validated external C-index across the grid.

Stage 2 — the active-factor subset, by ARD. Fit once at the chosen K_{init}; the normal prior on β (previous slide) lets Automatic Relevance Determination shrink unneeded factors' $\beta_k \to 0$ **on its own**, without a second per-K validation loop.

Stage 1: choosing K_{init}

- **ELBO** — the fitting objective itself, compared *across* K_{init}
- **BIC** $= -2\,\text{LL} + \log(n) \cdot \text{df}$, $\text{df} = K_{\text{init}} \cdot (n + p + 1)$ — an honest complexity penalty; **not** guaranteed to agree with ELBO or external C
- **Log-likelihood** — genomics + survival, an ELBO-style bound (not an exact marginal likelihood)
- **Cross-validated external C-index** — the only *generalization* signal among the four, evaluated on the five held-out cohorts
- These need **not** agree — see *Results* for the full $K_{\text{init}} = 2..20$ sweep

Stage 2: ARD classification at fixed K_{init}

Each factor k is classified by `classify_factors()`:

- **survival-active**: $|\hat{\beta}_k| > 0.001$

- **genomics-only**: not survival-active, but $\text{PVE}_k > 1\%$
- **dead**: neither — fully pruned

Running example: $K_{\text{init}} = 7$

7 starting factors → **2 survival-active** + **2 genomics-only** (4 kept, 3 pruned). Stable across a wide range of K_{init} — see *Results*.

Two stages. Stage 1 picks K_init by a consensus of ELBO, BIC, log-likelihood, and cross-validated external C-index — these don't have to agree, and the Results section shows the full sweep and where they disagree. Stage 2 is ARD: at a fixed K_init, the normal prior on beta lets survival-irrelevant factors shrink to beta=0 on their own, and classify_factors() splits the rest into survival-active (|beta|>0.001) vs genomics-only (PVE>1%) vs dead. K_init=7 is the running example throughout: 2 survival-active, 2 genomics-only, 4 kept total, 3 pruned — the same real-data configuration used throughout Results.

Results

Data — Synthetic Validation & PDAC Cohorts

Multi-cohort simulation

EBMF-grounded ground truth: shared factors (present in all studies) + study-specific factors; **shared factors carry the survival signal**, study-specific do not.

Scenarios

- *All Shared Factors* — every program is shared & prognostic
- *All Cohort-Specific Factors* — negative control; no shared signal ($\eta \equiv 0$)
- *Shared & Cohort-Specific Factors* — the realistic mix

Model Arms

- Unsupervised two-step (EBMF → Cox)
- The projection-predictor joint model, K selected by over-specified K_{init} + ARD pruning (mirroring the real-data procedure)

PDAC cohorts — External Validation

- **Train:** TCGA-PAAD + CPTAC ($n \approx 273$, RNA-seq + proteomics)
- **Validate:** 5 fully held-out cohorts, scored by the **exact** projection formula $\eta = (Y_{\text{new}} F)\beta$

Held-out cohort	Platform
Dijk	RNA-seq
Moffitt	Microarray
PACA-AU (array)	Microarray
PACA-AU (seq)	RNA-seq

Puleo	Microarray

Two datasets. The simulation now selects K the same way the real-data procedure does — an over-specified K_init pruned by ARD — instead of the oracle true-K used in June, and compares only the joint model against the unsupervised EBMF- then-Cox baseline. On real data we train on TCGA + CPTAC and validate on five fully held-out cohorts spanning microarray and RNA-seq, scored by the exact projection formula.

PDAC Cohorts — Prognostic Gene Programs

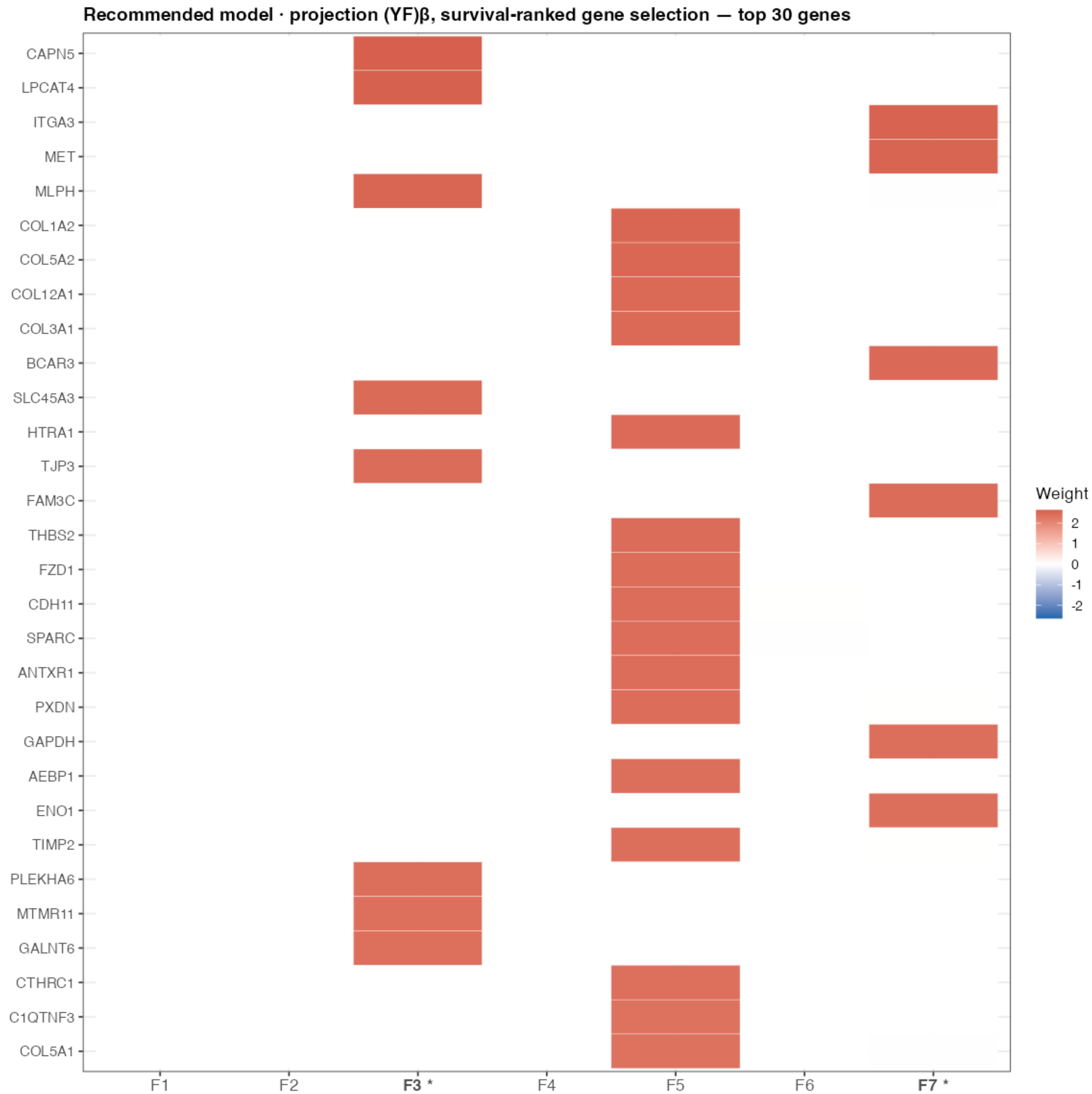

Figure 1: **Figure.** Gene-weight heatmap (top 30 genes) for the recommended model: projection $(YF)\beta$, per-platform z-standardization, survival-ranked gene selection (2064 genes), no cohort indicator, $K_{\text{init}} = 7$. Weights are **non-negative** (point-exponential prior). Survival-active programs ($|\hat{\beta}| > 10^{-3}$) are starred.

- $K_{\text{init}} = 7$ (ELBO / BIC / log-likelihood / external-C consensus — see the sweep below) → **2 survival-active** + 2 genomics-only kept — Program 3 (**protective**) and Program 7 (**adverse**)
- The *largest* expression program (collagen/ECM/stromal) is **not** survival-active — supervision separates expression-only structure from prognostic structure

On real data, the recommended fit starts at K_init=7 and ARD keeps 4 programs — 2 survival-active (Program 3 protective, Program 7 adverse) and 2 genomics-only. Note the largest expression program — the stromal collagen block — is NOT survival-active, exactly the separation supervision is supposed to give you. Weights are non-negative because L and F use a point-exponential prior.

PDAC Cohorts — Prognostic Association

The two survival-active programs stratify patients (split at median **projection score** $(YF)_k$ — the quantity the model scores on):

- **Adverse** — **Program 7** (higher activation → shorter survival), log-rank $p = 0.00047$
  - top genes: *ITGA3, MET, BCAR3, FAM3C, GAPDH, ENO1, TPI1, PGK1*
- **Protective** — **Program 3** (higher activation → longer survival), log-rank $p = 0.0077$
  - top genes: *CAPN5, LPCAT4, MLPH, SLC45A3, TJP3, PLEKHA6, MTMR11, GALNT6*

Proportional-hazards assumption not violated (Schoenfeld test).

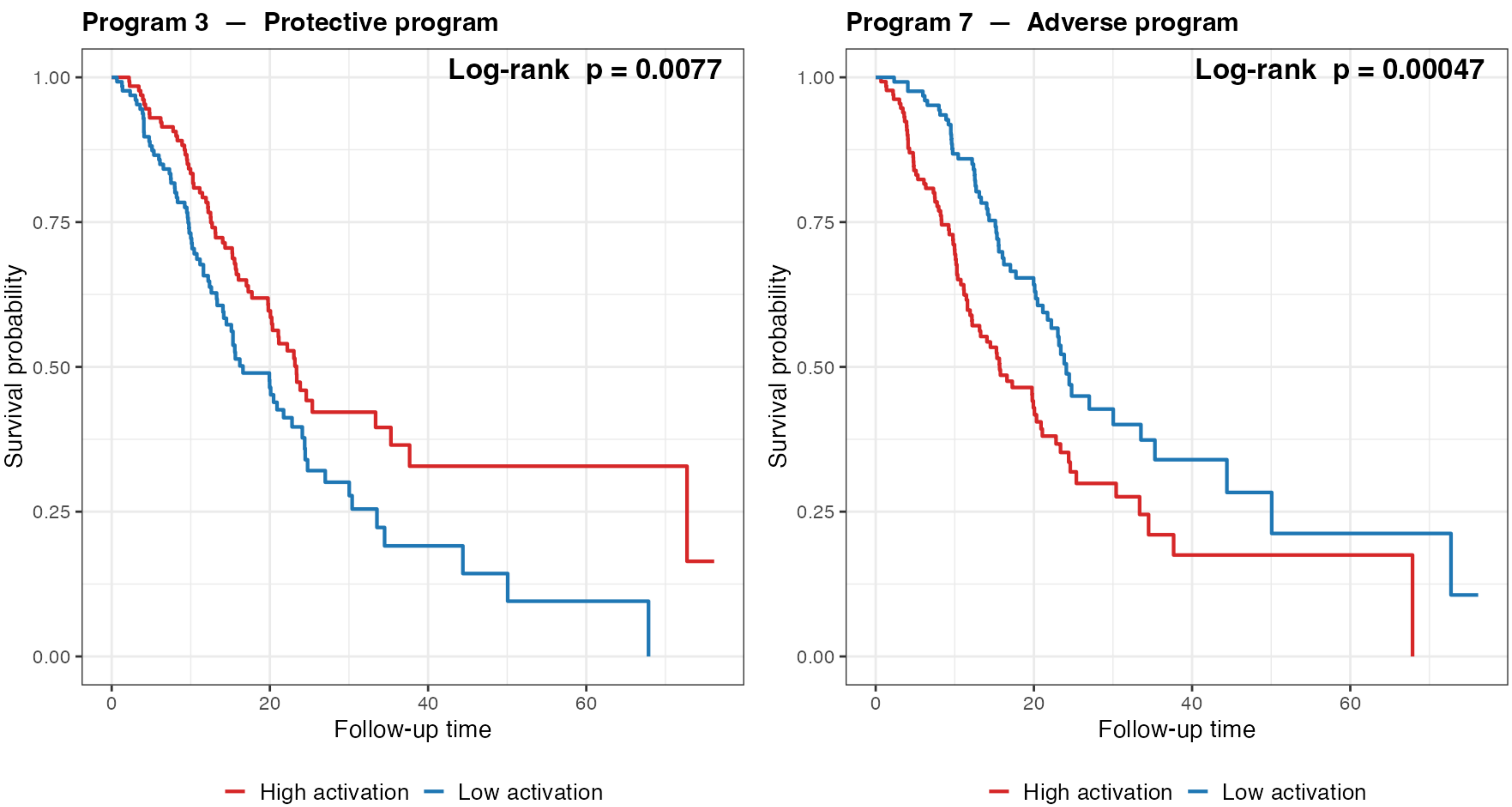

Figure 2: **Figure.** Kaplan–Meier curves for the two survival-active programs, patients split at the median projection score $(YF)_k$ in the training cohorts. The adverse program's high-activation group has worse survival; the protective program's, better. Direction read empirically from the curves.

The two active programs stratify patients into clear survival groups, split by the projection score (YF)__k. Program 7, the adverse program, carries MET, ITGA3, and glycolytic genes — biologically a classic aggressive signature. Program 3, protective, carries epithelial differentiation markers. Directions are read off the curves. Schoenfeld test shows PH holds.

Choosing K_{init}: the consensus sweep

$K_{\text{init}} = 2..20$, each fit classified by ARD. *(4-panel ELBO / BIC / log-likelihood / external-C figure inserted here — Phase 5.)*

- **ELBO-preferred:** $K_{\text{init}} = 3$
- **BIC-preferred:** $K_{\text{init}} = 3$
- **External-C-preferred:** $K_{\text{init}} = 12$

The three do **not** agree — reported honestly, not retuned to force a match. BIC's complexity penalty (~2,338 per unit K_{init}) dominates once K_{init} climbs past single digits, while external C-index keeps rewarding a bit more capacity before plateauing.

> Recommendation: $K_{\text{init}} = 7$
>
> Survival-active count is **stable at 2** across a wide range of K_{init} — added capacity beyond that goes to genomics-only or dead factors, ARD working as intended. $K_{\text{init}} = 7$ is kept as the practical default: reachable from a single fresh fit, externally competitive, and parsimonious ($K_{\text{eff}} = 4$ kept of 7).

Here is where the disagreement matters. ELBO and BIC agree with each other but pull toward a smaller K_init than external C-index does. I'm not going to pretend they agree — the honest read is that BIC's df penalty, which is on the order of tens of thousands per unit K here, dominates once K climbs into double digits, while external C-index keeps rewarding capacity a bit longer before plateauing. K_init=7 is kept as the practical recommendation because the survival-active count is stable there and beyond, and it's the simplest single-fit path — but I want to show you the actual curves and provoke the question of whether a different K_init in that plateau might be preferable.

Joint model vs. the unsupervised two-step

(Two-arm external C-index figure — projection $(YF)\beta$ vs. EBMF → Cox — inserted here, Phase 4.)

- **Projection model mean C = 0.627** · unsupervised EBMF → Cox 0.581, across the five held-out cohorts

- $K_{\text{init}} = 7$ (ELBO / BIC / log-likelihood / external-C consensus) → $K_{\text{eff}} = 2$ survival-active of 4 kept

The real-data analog of the synthetic slide, once the two-arm figure lands. Two arms: the recommended projection model, and the unsupervised EBMF-to-Cox two-step baseline.

Joint model vs. two-step as survival signal strength varies

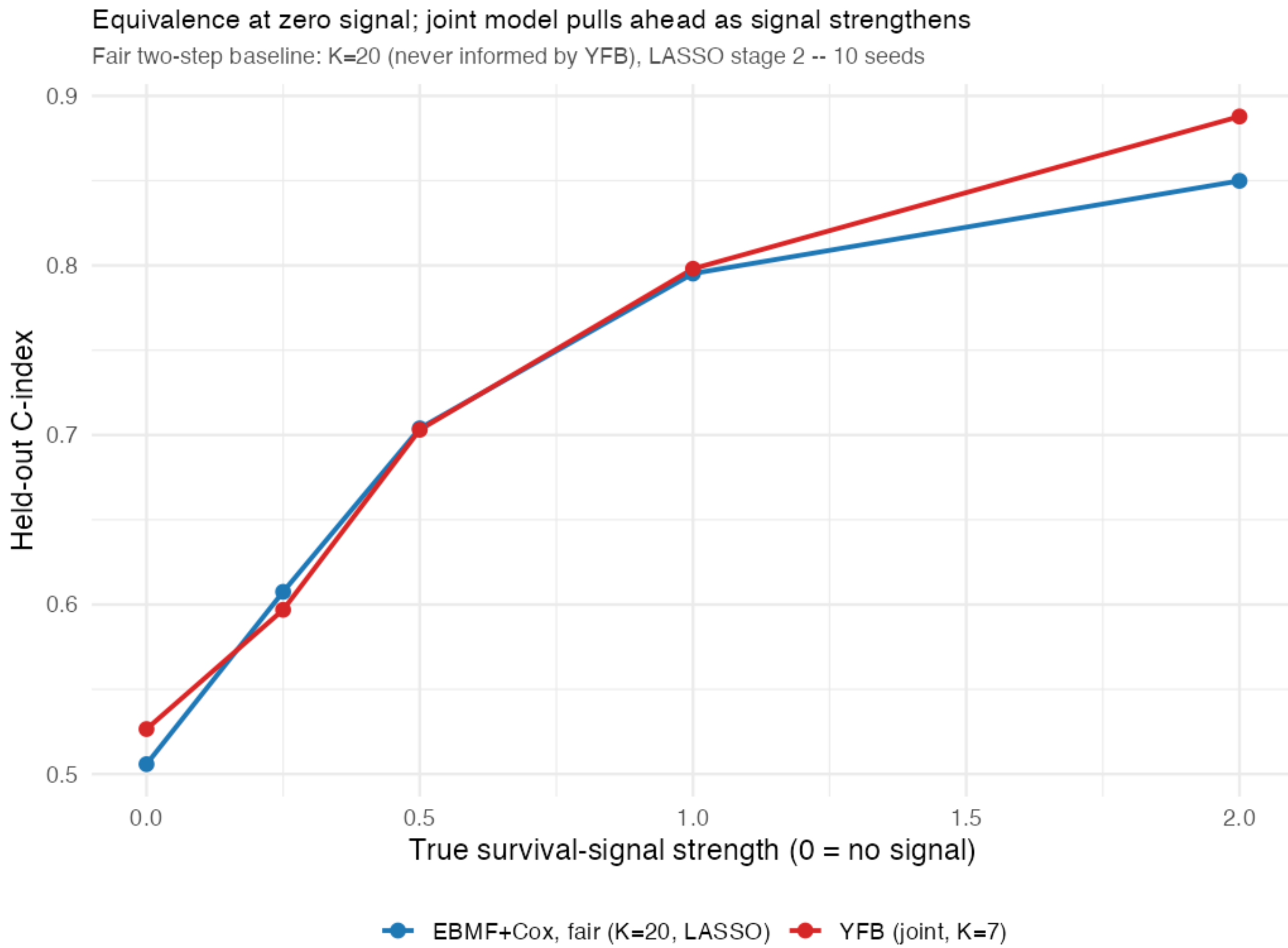

Figure 3: **Figure.** External performance of the joint model vs. the unsupervised two-step (EBMF → Cox) as the simulated survival-signal strength increases. At strength = 0 (no true survival signal in the data), the joint model matches EBMF — supervision costs nothing when there is nothing to exploit. At strength = 2, the joint model pulls ahead.

- **Strength = 0:** joint ≈ EBMF-only — the joint model does not fabricate prognostic signal when none exists
- **Strength = 2:** joint model outperforms the two-step — supervision helps *only when there is signal to find*, and does not hurt when there isn't

This single figure carries two points. At zero simulated survival signal, the joint model tracks EBMF almost exactly — no false discovery, no manufactured prognostic signal. As signal strength increases, the joint model pulls ahead of the two-step baseline. Together this is the cleanest argument that supervision is a free option: it costs nothing when uninformative and helps when it isn't.

Synthetic Results — Recovery & Held-out Performance

(ARD-based multi-cohort recovery simulation figures — recovery correlation, specificity accuracy, C-index by arm/scenario — inserted here or moved to the appendix, Phase 6, pending the placement decision.)

Placement pending: the simulation now selects K via over-specified K_init + ARD pruning instead of the June deck's oracle true-K. If ARD-based selection produces a materially new finding relative to June, this goes in the main body; if it essentially reproduces June under the new selection procedure, it moves to the appendix (never cut — it's still the evidence backing the method, and useful for Q&A).

Model Comparison Summary

Model	Linear predictor	Preprocessing
Projection (recommended)	**(YF)**	**per-platform z-std + survival-ra selection**
Unsupervised two-step	EBMF → Cox	per-platform z-std + survival-ranked g

Real ext. C = mean across 5 held-out PDAC cohorts. **K_init** = starting factor count, selected by an ELBO / BIC / log-likelihood / external-C consensus (Stage 1); **K_eff** = survival-active programs by ARD, $|\hat{\beta}| > 10^{-3}$ (Stage 2). Gene selection ranks genes by combined mean × variance per cohort.

- **Recommendation:** projection predictor $(YF)\beta$ · per-platform z-standardization · survival-ranked gene selection · **no** cohort indicator
- $K_{\text{init}} = 7 \to K_{\text{eff}} = 2$ survival-active (4 kept total), with **exact** external scoring

> **On gene selection**
>
> The combined-rank selection recipe was **motivated by** the lab's penalized survival-MF (DeSurv) preprocessing; here it is applied within the fully Bayesian model.

Two rows: the recommended projection model against the unsupervised two-step baseline. Projection predictor with per-platform z-std and survival-ranked gene selection, K_init=7 selected by consensus, is the recommendation.

Impact / key findings

1. **Joint supervision beats the unsupervised two-step**, and costs nothing when there is no survival signal to exploit (previous slide's signal-strength sweep).
2. **Preprocessing is decisive** — per-platform z-standardization before merging is the difference between a working model and $\beta \to 0$; **10 of 12** alternatives collapse.
3. K **selection is a two-stage split** — cross-validated external C-index (with ELBO, BIC, and log-likelihood) helps pick K_{init}; ARD shrinkage then determines the survival-active subset ($K_{\text{eff}} = 2$) directly from that single over-specified fit, with no separate per-K validation loop needed for K_{eff} itself.
4. **Mean external C-index = 0.627** across five independent PDAC cohorts — fully Bayesian throughout (adaptive priors, no α tuning, uncertainty-aware).

Limitations

- **BIC, ELBO, and external C-index need not agree** on K_{init} — reported honestly, not retuned to force a match (previous slides)
- **Small external cohorts give noisy estimates** — per-cohort C ranges down to 0.55 (Moffitt), near chance
- The active programs are **statistically** prognostic and **biologically annotated** (pathway enrichment: Program 7 basal-like/adverse, Program 3 classical/protective, confirmed by 5 independent methods) — a matched head-to-head against DeSurv on one protocol remains a planned next step

Four takeaways, then an honest limitations beat: the K-selection criteria disagree with each other and that's reported as a finding, not smoothed over; small external cohorts give noisy per-cohort estimates; and the biological annotation of the two active programs is done, with a matched DeSurv comparison still on the to-do list.

Future directions & summary

Current status

- K **selection**: cross-validated external C-index, ELBO, BIC, and log-likelihood inform K_{init}; ARD shrinkage determines K_{eff}
- **Gene programs biologically annotated** — pathway enrichment on both survival-active programs, 5 independent methods

Next steps

1. **Multiple cohorts / multiple modalities via indicator columns in** L — stack methylation + expression, possibly with modality-specific priors; leave-one-study-out validation; train on A vs. B vs. A+B
2. **Manuscript preparation** — methods-primary framing; PDAC as the application

Summary

1. Joint supervised factorization via the projection predictor $(YF)\beta$; **point-exponential priors on L, F, a normal prior on β**, non-parametric baseline hazard
2. **Preprocessing is crucial** — per-platform normalization prevents the survival precision from being dominated and β from collapsing
3. K **selection: cross-validated external C-index, ELBO, BIC, and log-likelihood inform $K_{\mathbf{init}}$; ARD shrinkage determines $K_{\mathbf{eff}}$** — validated on real data (five external PDAC cohorts, mean C = 0.627) and synthetic data with known ground truth
4. **Recommended model**

- Linear predictor: projection $(YF)\beta$
- L cohort indicator: No (candidate direction for multi-cohort/multi-modality extension, above)
- Pre-processing: per-platform z-std
- $K_{\text{init}} = 7 \rightarrow K_{\text{eff}} = 2$, mean external C = **0.627**

K selection: cross-validated external C-index, ELBO, BIC, and log-likelihood inform K_init; ARD shrinkage determines K_eff. Both survival-active programs are now biologically annotated. The forward-looking direction I want to lead with is extending L with indicator columns for multiple cohorts and/or multiple modalities — stack methylation and expression, possibly with modality-specific priors, and validate with leave-one-study-out and train-on-A-vs-B-vs-A+B comparisons. Then manuscript prep.

Discussion

Thank You

Questions, suggestions, and feedback welcome!

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Open the floor. Appendix slides cover convergence diagnostics, survival-activity magnitudes, and the DeSurv comparison.

Appendix

Appendix — convergence & survival-activity magnitudes

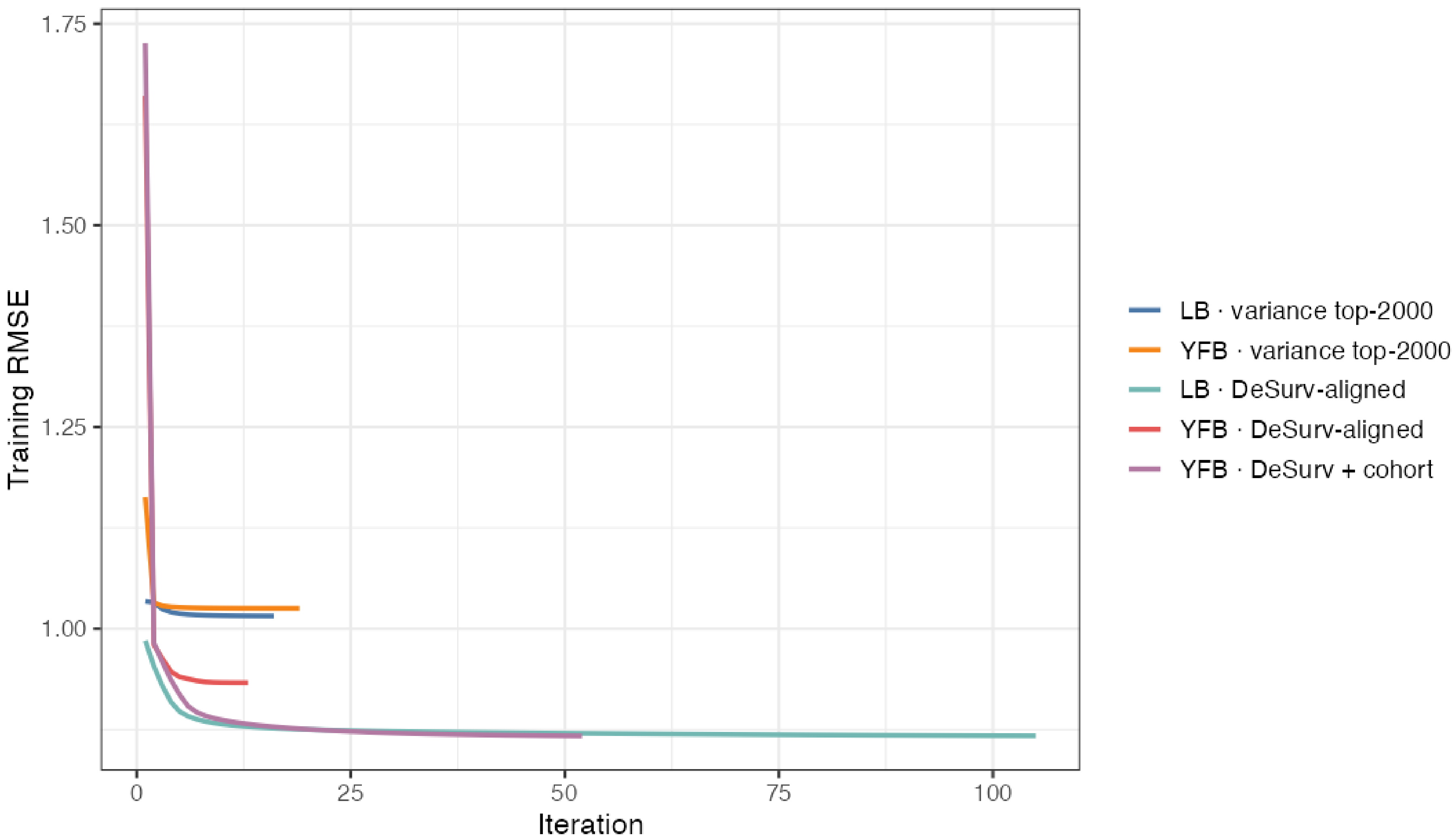

Figure 4: **Figure A1.** Training RMSE convergence traces across model configurations — RMSE stabilizes to a plateau; stopping is governed by the relative ELBO change falling below 10^{-5} ($\Delta L, \Delta\beta$ tracked as diagnostics).

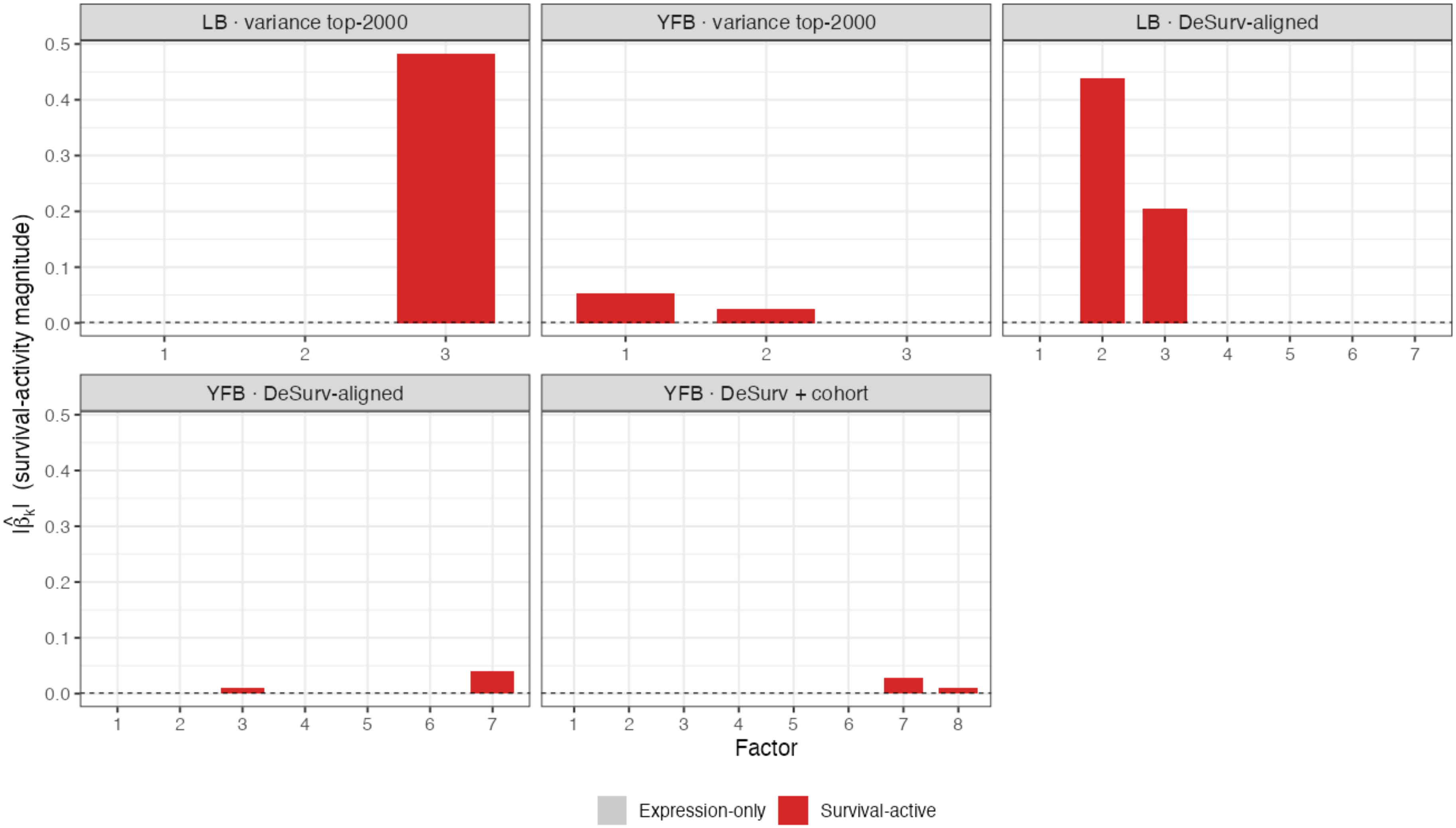

Figure 5: **Figure A2.** Survival-activity magnitude $|\hat{\beta}_k|$ per program. Most programs fall below the activity threshold (expression-only); a small number clear it (survival-active).

- The spike-and-slab / shrinkage behavior is visible: most programs are shrunk below the survival-activity threshold, with only the survival-active ones retained

Appendix — relation to DeSurv (Young et al., PNAS 2026)

Same design & same data — SBMF is the Bayesian counterpart. Both jointly fit supervised NMF + Cox with **non-negative gene programs**, share the same training set (**TCGA + CPTAC,** $n = 273$) and the **same five external PDAC cohorts** (Dijk, Moffitt, PACA-AU array/seq, Puleo; $n = 616$), score new patients by **projection** (no retraining), and use the **combined-rank top-3000-per-cohort** gene selection. Both also find that the **highest-variance program is *not* the most prognostic**.

DeSurv objective: $(1-\alpha)\,\mathcal{L}_{\mathrm{NMF}} - \alpha\,\mathcal{L}_{\mathrm{Cox}}$, with the survival gradient on the gene programs W.

What SBMF changes

	DeSurv	SBMF
Estimation	penalized optimization	Bayesian variational (CAVI)
K selection	Bayesian-optimization over k, α, λ jointly	ARD shrinkage + ELBO/BIC/log-likelihood/external-C consensus on K_{init} only
Shrinkage	fixed elastic-net	empirical-Bayes priors (learned)
Uncertainty	point estimates	posterior mean + variance
Noise	least-squares	per-gene precision τ; $E[L^2]$ correction

- **DeSurv:** $k=3,\ \alpha=0.334$; pooled external **HR/SD** $=1.50$ (95% CI 1.31–1.72)
- **SBMF (YF) :** $K_{\mathrm{init}}=7 \to K_{\mathrm{eff}}=2$; mean external **C $=0.627**

Honest caveat: the two report different primary metrics (DeSurv: pooled HR/SD; SBMF: mean external C-index) and use slightly different normalization (DeSurv within-subject rank → 1970 genes; SBMF per-platform z-std → 2064). A **matched same-protocol head-to-head is a planned next step** — DeSurv is *not* re-run here.

This is the backup for the inevitable "how does this compare to DeSurv?" question. SBMF is the Bayesian sibling of DeSurv — same supervised NMF+Cox idea, same training data and the same five external cohorts, same projection-based scoring, and SBMF borrowed DeSurv's combined-rank gene selection. The genuine differences: DeSurv tunes k, alpha, and lambda jointly by Bayesian optimization; SBMF selects K_init by a consensus of fit/prediction metrics and lets ARD prune within that fit, with empirical-Bayes priors learning the shrinkage and no alpha or lambda to tune, plus posterior uncertainty and a heteroscedastic noise model. On numbers, DeSurv reports pooled HR per SD of 1.50; SBMF reports mean external C of 0.627 — different metrics, so a matched head-to-head on one protocol is the honest next step, not a claim of superiority today.